WORKED SOLUTION

STEP II 2013, Question 5

Pure • Symmetry and Curve Sketching

STEP II 2013, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q5

The key idea

If a function is unchanged by a substitution such as $x \mapsto 1-x$ or $x \mapsto 1/x$, then differentiating the identity with the chain rule ties the gradient at x to the gradient at the image point. At a point the substitution leaves fixed, the gradient must equal its own negative, so it is zero. For the equations in part (iii), the symmetry suggests writing everything in terms of $u = x^2 - x$, which is itself unchanged by $x \mapsto 1-x$; a cubic in u is then much easier to handle than a sextic in x.

Solution

Part (i)

Differentiating $\mathrm{f}(x) = \mathrm{f}(1-x)$ with respect to x, using the chain rule on the right,

$$\mathrm{f}'(x) = -\mathrm{f}'(1-x) \qquad \text{for all } x.$$

Putting $x = \frac{1}{2}$ gives $\mathrm{f}'(\frac{1}{2}) = -\mathrm{f}'(\frac{1}{2})$, so $\mathrm{f}'(\frac{1}{2}) = 0$.

Now suppose also that $\mathrm{f}(x) = \mathrm{f}(1/x)$ for $x \neq 0$. Differentiating, since $\frac{\mathrm{d}}{\mathrm{d}x}(\frac{1}{x}) = -\frac{1}{x^2}$,

$$\mathrm{f}'(x) = -\frac{1}{x^2}\,\mathrm{f}'\left(\frac{1}{x}\right) \qquad (x \neq 0).$$

At $x = -1$ (which is its own reciprocal) this reads $\mathrm{f}'(-1) = -\mathrm{f}'(-1)$, so $\mathrm{f}'(-1) = 0$.

At $x = 2$ it gives $\mathrm{f}'(2) = -\frac{1}{4}\mathrm{f}'(\frac{1}{2}) = 0$, by the first result. (Equally, the first identity with $x = 2$ gives $\mathrm{f}'(2) = -\mathrm{f}'(-1) = 0$.)

Part (ii)

Replace x by $1/x$. The numerator becomes

$$\left(\frac{1}{x^2} - \frac{1}{x} + 1\right)^3 = \frac{(1 - x + x^2)^3}{x^6},$$

and the denominator becomes

$$\left(\frac{1}{x^2}-\frac{1}{x}\right)^2=\frac{(1-x)^2}{x^4}.$$

Dividing, and using $x^2(1-x)^2=(x^2-x)^2$,

$$\mathrm{f}\left(\frac{1}{x}\right)=\frac{(x^2-x+1)^3}{x^6}\cdot\frac{x^4}{(1-x)^2}=\frac{(x^2-x+1)^3}{x^2(x-1)^2}=\mathrm{f}(x).$$

Replace x by $1-x$: we have $(1-x)^2-(1-x)=(1-x)(-x)=x^2-x$, so x^2-x is unchanged, and hence so is x^2-x+1. Therefore $\mathrm{f}(1-x)=\mathrm{f}(x)$.

The sketch. By part (i), $x=-1$, $\frac{1}{2}$ and 2 are stationary points; since there are exactly three, these are all of them. Their values are

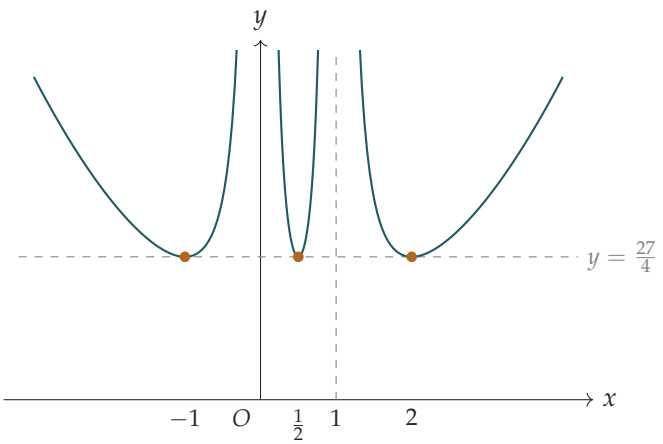
$$\mathrm{f}(-1)=\frac{3^3}{2^2}=\frac{27}{4},\qquad \mathrm{f}(\tfrac{1}{2})=\frac{(3/4)^3}{(1/4)^2}=\frac{27}{4},\qquad \mathrm{f}(2)=\frac{3^3}{2^2}=\frac{27}{4}.$$

All three are the same, as they must be: $\mathrm{f}(2)=\mathrm{f}(\frac{1}{2})$ and $\mathrm{f}(2)=\mathrm{f}(1-2)=\mathrm{f}(-1)$.

Other features:

- $x^2-x+1=(x-\frac{1}{2})^2+\frac{3}{4}>0$, so $\mathrm{f}(x)>0$ everywhere; the curve never meets the x-axis.
- As $x\to 0$ or $x\to 1$ the numerator tends to 1 and the denominator to 0^+, so $\mathrm{f}(x)\to+\infty$ from both sides: $x=0$ and $x=1$ are vertical asymptotes.
- For large $|x|$, $\mathrm{f}(x)\approx x^6/x^4=x^2$, so $\mathrm{f}(x)\to+\infty$ as $x\to\pm\infty$, roughly like a parabola.

So on each of the three intervals $(-\infty,0)$, $(0,1)$, $(1,\infty)$ the function is continuous and tends to $+\infty$ at both ends. Each interval must therefore contain a minimum, which is a stationary point; each contains exactly one of -1, $\frac{1}{2}$, 2, so that point is the minimum and there is no other turning point. Each branch is U-shaped with lowest value $\frac{27}{4}$.



Part (iii)

The equation $f(x) = \frac{27}{4}$**.** From the sketch, $\frac{27}{4}$ is the minimum value on every branch and is attained only at the three stationary points. So the roots are

$$x = -1, \qquad x = \tfrac{1}{2}, \qquad x = 2,$$

and $f(x) > \frac{27}{4}$ for every other x in the domain, that is for

$$x < -1, \quad -1 < x < 0, \quad 0 < x < \tfrac{1}{2}, \quad \tfrac{1}{2} < x < 1, \quad 1 < x < 2, \quad x > 2.$$

We can confirm this algebraically. Put $u = x^2 - x$ (so $u \neq 0$ in the domain); then $f(x) = \dfrac{(u+1)^3}{u^2}$ and

$$f(x) - \frac{27}{4} = \frac{4u^3 - 15u^2 + 12u + 4}{4u^2} = \frac{(u-2)^2(4u+1)}{4u^2}.$$

Since $4u + 1 = 4x^2 - 4x + 1 = (2x-1)^2 \geq 0$, this is never negative, and it is zero exactly when $u = 2$ (that is, $x^2 - x - 2 = 0$, so $x = 2$ or -1) or $x = \frac{1}{2}$.

Ask yourself How can we guess a root of $f(x) = \frac{343}{36}$ without solving a degree-six equation?

The equation $f(x) = \frac{343}{36}$**.** Note $\frac{343}{36} = \frac{7^3}{6^2}$, so we look for x with $x^2 - x + 1 = 7$ and $x^2 - x = 6$. Both hold for $x = 3$, so $f(3) = \frac{343}{36}$. The symmetries now hand us more roots for free: applying $x \mapsto 1 - x$ and $x \mapsto 1/x$ repeatedly to 3 gives

$$3, \quad 1 - 3 = -2, \quad \tfrac{1}{3}, \quad 1 - \tfrac{1}{3} = \tfrac{2}{3}, \quad \tfrac{3}{2}, \quad 1 - \tfrac{3}{2} = -\tfrac{1}{2}.$$

These are six distinct numbers, and clearing denominators turns $f(x) = \frac{343}{36}$ into a polynomial equation of degree six, so these are all the roots:

$$x = -2, \ -\tfrac{1}{2}, \ \tfrac{1}{3}, \ \tfrac{2}{3}, \ \tfrac{3}{2}, \ 3.$$

(In terms of u, the equation is $36u^3 - 235u^2 + 108u + 36 = 0$, which factorises as $(u-6)(9u+2)(4u-3) = 0$, giving the pairs $\{3, -2\}$, $\{\frac{1}{3}, \frac{2}{3}\}$ and $\{\frac{3}{2}, -\frac{1}{2}\}$.)

Since $\frac{343}{36} > \frac{27}{4}$, the horizontal line $y = \frac{343}{36}$ cuts each U-shaped branch twice, once on each side of its minimum, and the branch lies above the line outside these two crossings. The branch on $(-\infty, 0)$ is cut at -2 and $-\frac{1}{2}$, the one on $(0, 1)$ at $\frac{1}{3}$ and $\frac{2}{3}$, and the one on $(1, \infty)$ at $\frac{3}{2}$ and 3. Hence $f(x) > \frac{343}{36}$ exactly when

$$x < -2, \quad -\tfrac{1}{2} < x < 0, \quad 0 < x < \tfrac{1}{3}, \quad \tfrac{2}{3} < x < 1, \quad 1 < x < \tfrac{3}{2}, \quad x > 3.$$

Checks

- Try a value: $f(-\frac{1}{2}) = \dfrac{(7/4)^3}{(3/4)^2} = \dfrac{343/64}{9/16} = \dfrac{343}{36}$, as claimed.

- A point inside a claimed interval: $\mathrm{f}(4) = \frac{13^3}{12^2} \approx 15.3 > \frac{343}{36} \approx 9.53$, while $\mathrm{f}(2) = 6.75$ lies below, consistent with $(\frac{3}{2}, 3)$ being excluded.
- The six roots come in pairs summing to 1 and in reciprocal pairs, exactly as the two symmetries require; a computer check confirmed the inequalities at 10^5 random points.

Common slips

- Forgetting the factor $-1/x^2$ from the chain rule when differentiating $\mathrm{f}(1/x)$. It doesn't matter at $x = -1$, but it is essential to see why the gradient at 2 vanishes.
- Writing the ranges as single intervals such as $x < -2$ or $x > 3$, and forgetting that the asymptotes at $x = 0$ and $x = 1$ split the solution set; $x = 0$ and $x = 1$ must be excluded.
- Treating the stationary points as maxima. Every branch tends to $+\infty$ at both ends, so they are minima.

Beyond STEP

The six maps generated by $x \mapsto 1 - x$ and $x \mapsto 1/x$ form a group (isomorphic to the symmetries of a triangle), and f is, up to a constant factor, the classical j-invariant of the cross-ratio: the function that is unchanged by all six. Functions invariant under a group of substitutions are a central theme of Galois theory and of the theory of elliptic curves. Allenby's *Rings, Fields and Groups* is a good place to meet groups of transformations like this one.

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